Designing a stable energy landscape

Stable de novo protein design via joint conformational landscape and sequence optimization

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Can two imperfect objectives make more stable proteins?

- Inverse folding asks: which sequence fits this backbone?
- Structure prediction asks: which backbone fits this sequence?
- The design goal is stronger: a sequence with a stable, preferred folded state.

Central hypothesis
Jointly optimizing sequence and predicted structure can improve stability beyond optimizing either conditional model alone.

Introduction; Fig. 1. The landscape argument is the authors' interpretation, not a directly measured free-energy surface.

First separate probability, confidence, and thermodynamics

Quantity	Meaning in this study
$P(S \mid X)$	Sequence compatibility with structure X; TrMRF / ProteinMPNN.
$P(X \mid S)$	Structural distribution given sequence S; TrRosetta (TrROS).
pLDDT	Structure model confidence; useful predictor, not folding free energy.
$\Delta G_{\mathrm{unfold}}$	Experimental stability estimate; larger positive values mean more stable folding.

Introduction; Methods; Fig. 2. Model probabilities and confidence scores are proxies, not calibrated physical energies.

Why either direction can fail alone

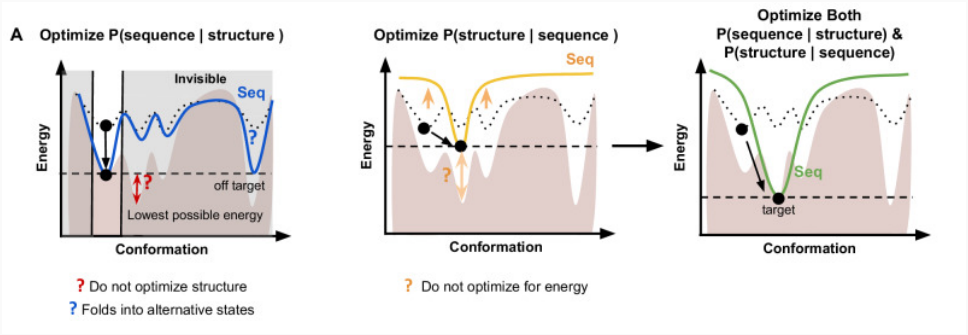


Fig. 1A (author schematic): fixed-backbone compatibility may miss competing folds; structure prediction may accept adversarial sequences.

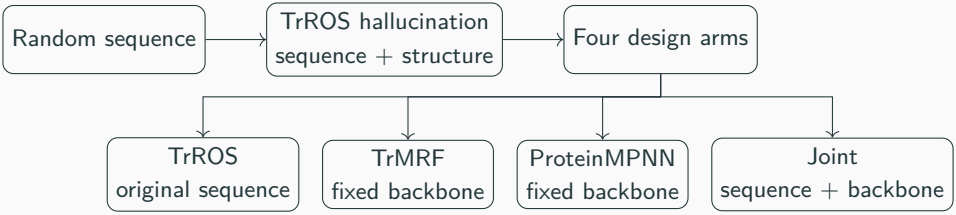
A concrete example: redesigning a small beta fold

1. Start with a hypothetical 60-residue sequence and obtain a compact beta-rich backbone.
2. Fixed-backbone redesign may improve local sequence compatibility while another strand arrangement remains favorable.
3. Joint optimization changes sequence *and* predicted distances/orientations; the backbone can accommodate the new side chains.

Desired outcome
Retain the overall fold, improve stability, and avoid rewarding a sequence merely because one model is confident.

Illustrative example, not an experimentally tested sequence. Actual designs: fewer than 80 residues; Supplementary Figs. 1–3.

The comparison uses four sequences per starting fold



The matched-fold comparison reduces structural differences while asking how sequence generation changes stability.

Results; Fig. 1B; Supplementary Fig. 2. Joint backbones may undergo modest perturbations.

TrROS creates structure; TrMRF reads it in reverse

TrROS

- Sequence features $\rightarrow$ distance/orientation distributions.
- Backpropagate to sequence from a KL objective relative to background geometry.
- Rosetta reconstructs a 3D model from predicted 6D features.

TrMRF

- Structural bins $\rightarrow$ conservation and pairwise couplings.
- Train by sequence cross-entropy / MSA pseudo-likelihood.
- 15,051 nonredundant PDB proteins; released before May 2018.

Methods: TrROS, TrMRF, and joint models. TrMRF training set filtered at $\leq$ 30% sequence identity.

Joint optimization updates inputs, not model weights

$$S \xrightarrow{\text{TrROS}} \text{6D geometry} \xrightarrow{\text{TrMRF}} \text{sequence compatibility}$$

- Initialize from the final TrROS-generated sequence.
- Combine the TrROS and TrMRF loss signals; backpropagate to the input sequence.
- Use discrete sequence samples with a straight-through gradient estimator.
- Continue until loss no longer decreases; the predicted geometry changes with the sequence.

What is joint here?
Two pretrained directional models are coupled at design time; this is not a newly trained joint generative model.

Methods; Supplementary Fig. 2. ProteinMPNN comparator uses vanilla model, sampling temperature 0.1.

Experimental design: many proteins, one stability readout

- 5,167 starting designs $\times$ 4 methods $=$ 20,668 measured sequences.
- Mini-proteins below 80 residues; no cysteine to avoid disulfide stabilization.
- cDNA display proteolysis with trypsin and chymotrypsin estimates $\Delta G_{\mathrm{unfold}}$.
- AF2 dimer predictions remove potential oligomer/aggregation artifacts: 13,442 sequences retained.

Most controlled subset
1,427 complete four-method groups $=$ 5,708 sequences with matched starting folds.

Results; Methods: Data filtration. Low inter-chain PAE suggests potential association; typical monomer criterion inter-PAE > 15.

How protease resistance becomes a folding free energy

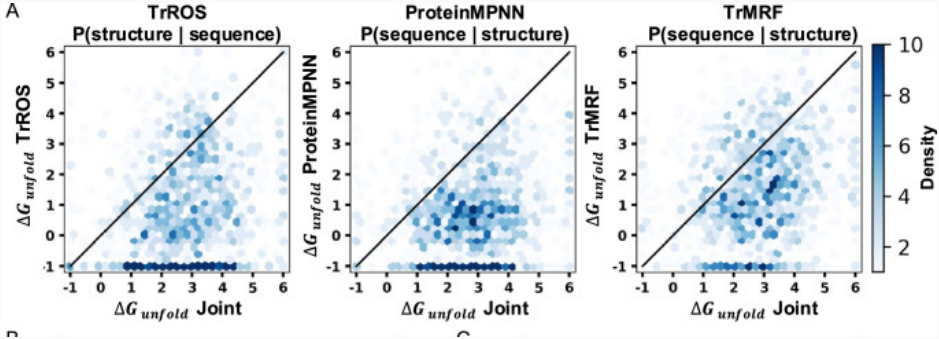
- Sequence counts after proteolysis across 11 concentrations plus no-protease control yield K_{50} by Bayesian inference.
- A sequence-specific unfolded-state model estimates $K_{50,U}$; folded-state cleavage $K_{50,F}$ is assumed constant across constructs.

$$\frac{[U]}{[F]} \approx \frac{1/K_{50} - 1/K_{50,F}}{1/K_{50,U} - 1/K_{50}}, \qquad \Delta G_{\mathrm{unfold}} = -RT \ln \frac{[U]}{[F]}.$$

Key assumptions
Small proteins approximate a two-state system; cleavage kinetics and the unfolded-state susceptibility model connect resistance to stability. Two proteases provide a consistency check.

Supplementary Methods A, Eqs. 4–7; Supplementary Figs. 6–7. Protease concentrations form a threefold dilution series.

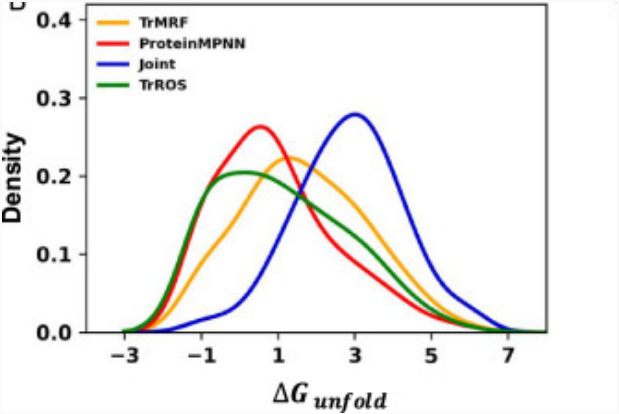
Within the same fold, joint designs usually win



Joint has greater ΔG_{unfold} than TrROS in **80.5%**, TrMRF in **74.4%**, and ProteinMPNN in **84.7%** of matched comparisons.

Fig. 2A; Results. Axes are kcal/mol. Below the diagonal favors joint; matched analysis uses 1,427 folds.

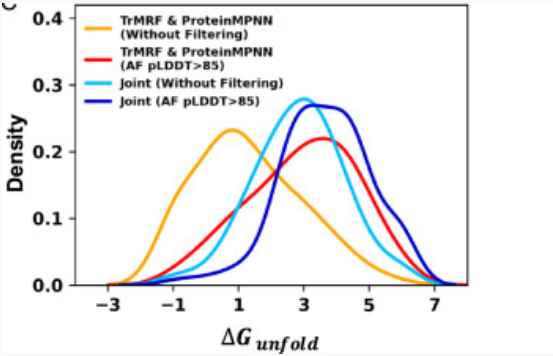
The whole stability distribution shifts



The joint arm shifts toward higher stability, rather than producing only a few exceptional hits. This supports the practical design benefit in this mini-protein distribution. It does not directly demonstrate a global thermodynamic minimum.

Fig. 2B; interpretation distinguishes measured stability from the hypothesized landscape mechanism.

Confidence filtering trades yield for stability



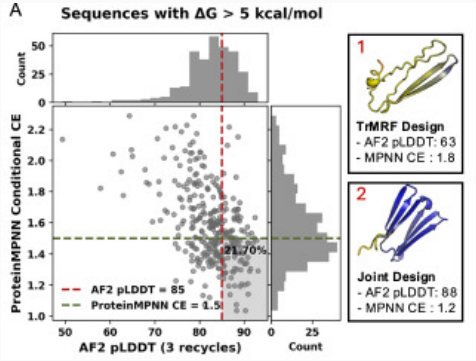
Design set	Retained	Median
Inverse models	237/2,854	3.08
Joint, unfiltered	100%	2.90
Joint, filtered	12%	3.66

Medians: kcal/mol. Filter: AF2 pLDDT > 85.

Joint generation improves the starting distribution; filtering can still improve it further.

Fig. 2C; Results. Inverse models pooled: TrMRF + ProteinMPNN; common complete-fold comparison set.

Stable proteins can fail standard computational filters



Among designs with $\Delta G_{\text{unfold}} > 5$ kcal/mol, only **21.7%** pass both:

- AF2 pLDDT > 85;
- ProteinMPNN cross-entropy < 1.5.

Confidence thresholds can enrich stable designs while discarding many stable proteins.

Fig. 3A; Results. This is recall among highly stable proteins, not the precision of the accepted set.

Two ranking questions reveal different model strengths

- **Across folds:** can a score rank the absolute stability of structurally different designs?
- **Within a fold:** can score differences rank sequence changes around the same starting backbone?

AF2 / ESMFold confidence performs well for absolute stability. ESM-2 sequence pseudo-likelihood performs poorly there, yet becomes informative for differences within folds.

Interpretation
Sequence plausibility and structural compatibility contribute complementary information; one metric need not solve both tasks.

Fig. 3B. Spearman rank correlation over 13,442 sequences; within-fold comparisons use common starting structures.

Prediction settings change stability correlation

- Fewer AF2 recycles improve stability correlation relative to the standard three-recycle setting.
- Fully masking the AF2 MSA input and averaging eight seeds improves reported Spearman correlation from 0.50 to 0.54.
- ESMFold benefits from averaging eight predictions with 50% input masking.
- Boltz-2 does not improve the stability correlation over AF2 / ESMFold in this evaluation.

Proposed explanation
Refinement and evolutionary priors can give high confidence to unstable artificial sequences; masking may expose uncertainty. This is a hypothesis, not a universal rule.

Results; Supplementary Fig. 9. These are stability-ranking settings, not recommendations for highest structural accuracy.

Hybrid scores combine complementary views

Let C be ProteinMPNN conditional cross-entropy, E ESM-2 pseudo-likelihood, F half-masked ESMFold pLDDT, and A fully masked AF2-MSA pLDDT.

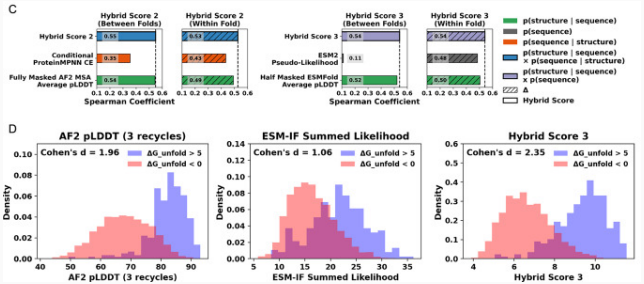
$$H_1 = -0.90C + 0.10F$$
$$H_2 = -0.90C + 0.10A$$
$$H_3 = \quad 0.86E + 0.14F$$

Weights selected using 20% of the data; final correlations evaluated on the remaining 80%.

An important qualification
Base model scores are zero-shot, but the mixture weights use experimental stability labels. These hybrids are calibrated predictors.

Results: hybrid-score definitions. Higher H predicts greater stability; formulas shown as reported.

Hybrid ranking improves separation in this dataset



Hybrid 2: $\rho = 0.55$ across folds, 0.53 within folds. Hybrid 3: 0.54 for both.

Fig. 3C–D. Stable: $\Delta G_{\text{unfold}} > 5$; unstable: < 0 kcal/mol. Hybrid 2 and 3 achieve the strongest reported correlations.

What sequence features change?

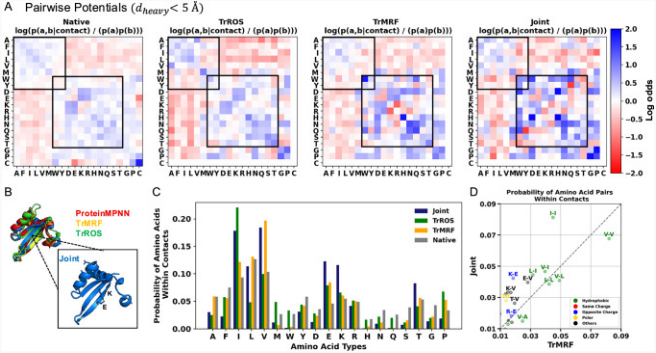


Fig. 4. Contact: side-chain heavy atoms < 5 Å, separated by > 5 residues; AF2-predicted structures; 200 native PDB controls.

Hydrophilic contacts suggest a mechanism, not proof

- Joint designs enrich polar/charged contact pairs; lysine–glutamate contacts are prominent.
- Predicted structures show possible electrostatic interactions and hydrogen bonds.
- Polar contacts tend to be more buried than in TrROS designs; charged pairs remain surface exposed.

Evidence boundary
Contact statistics and predicted side-chain geometries are consistent with stabilization. Targeted mutations or experimental structures would be needed to establish causal contributions.

Fig. 4; Supplementary Fig. 15. The paper does not directly measure per-contact free energies or the full conformational landscape.

What can we conclude, and what remains open?

1. **Established here:** joint TrROS–TrMRF design improves measured stability for small, mostly beta-containing de novo proteins.
2. **Scope:** fewer than 80 residues, no cysteine; 54% beta, 32% alpha+beta, 8% alpha. Starting folds come from TrROS.
3. **Assay/structure limits:** proteolysis-derived estimates and computational oligomer filters; predicted folds do not establish experimental target structures.
4. **Generalization:** larger proteins, functions, different backbone generators, and fold-disjoint calibration remain useful tests.

Discussion; Supplementary Fig. 5; Methods. Items framed as future tests are journal-club interpretation.

Discussion: optimize a generator or screen more candidates?

Take-home result
Coupling sequence and structure models improves the stability distribution; their scores are also complementary for ranking designs.

1. At a fixed compute budget, when does joint optimization beat cheap inverse folding followed by screening?
2. Would gains persist with modern backbones and a fold-held-out test set?
3. Which experiment would distinguish better target stabilization from avoiding alternative structures?

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